

7.3 Halogen Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.3 Halogen Compounds (A Level Only)
Difficulty	Easy

Time allowed: 20

Score: /15

Percentage: /100

Question 1a

At room temperature and pressure, bromine reacts with benzene, in the presence of a halogen carrier to produce bromobenzene. This is an electrophilic substitution reaction.

The electrophile is the bromonium ion, Br^+ and is generated when bromine reacts with the halogen carrier, e.g. AlBr_3 .

Complete the mechanism in Fig. 1.1 to show how the bromonium ion is produced by adding any full and partial charges and curly arrows.



Fig. 1.1

[2 marks]

Question 1b

Complete the mechanism in Fig. 1.2 to show how bromobenzene is produced from Br^+ and benzene. Include any relevant charges and curly arrows.

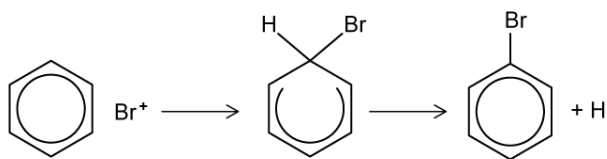


Fig. 1.2

[3 marks]

Question 1c

Write an equation to show the regeneration of the catalyst.

[1 mark]

Question 2a

Methylbenzene can react with chlorine under different conditions to give the monochloro derivatives **F** and **G**, shown in Fig. 2.1.

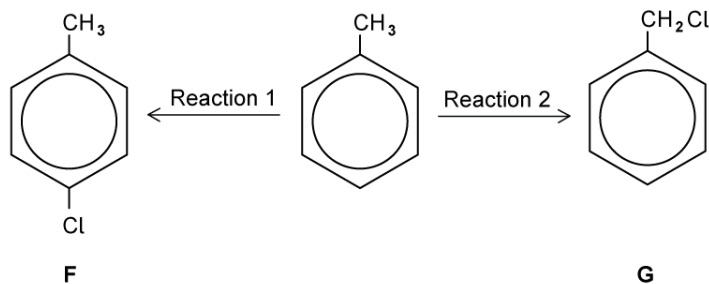


Fig. 2.1

Suggest reagents and conditions for each reaction.

reaction 1

reaction 2

[2 marks]

Question 2b

Identify the type of reaction mechanism that occurs in each reaction in Fig. 2.1.

reaction 1

reaction 2

[2 marks]

Question 2c

When the reagent used in reaction 1 is present in excess, further organic products are formed.

Draw the structure of one of these organic products and give its name.

Structure

Name

[2 marks]

Question 3a

2-bromopentane is a halogenoalkane and bromobenzene is a halogenoarene.

State which is the most reactive compound.

[1 mark]

Question 3b

The average bond enthalpy of a C–Br bond is 276 kJ mol^{-1} . Suggest whether the bond enthalpy of the C–Br bond in bromobenzene is greater or less than the average bond enthalpy and state what this means about the strength of the C–Br bond in bromobenzene.

[1 mark]

Question 3c

2-bromopentane readily undergoes nucleophilic substitution with a hydroxide ion, OH^- , to form pentan-2-ol whereas bromobenzene requires extreme conditions to react when attacked by a nucleophile, such as OH^- .

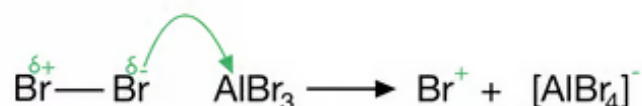
Suggest what happens, under standard conditions, as a hydroxide ion approaches a molecule of bromobenzene.

[1 mark]

Model Answers: Easy

1a

a) The completed mechanism to show how the bromonium ion is produced is:



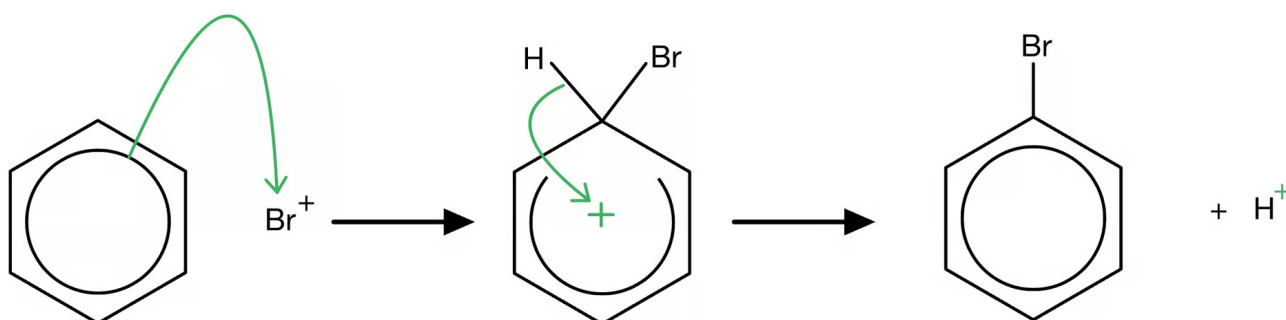
- Correct partial charges on Br_2 and full charges on Br and $[\text{AlBr}_4]$; [1 mark]
- Curly arrow going from δ^- Br to Al; [1 mark]

[Total: 2 marks]

- A lone pair of electrons of one of the bromine atoms in the bromine molecule is donated to AlBr_3 , forming a coordinate bond
- Electrons are drawn from the other bromine atom in the bromine molecule

1b

b) The completed mechanism to show how bromobenzene is produced from Br^+ and benzene is:

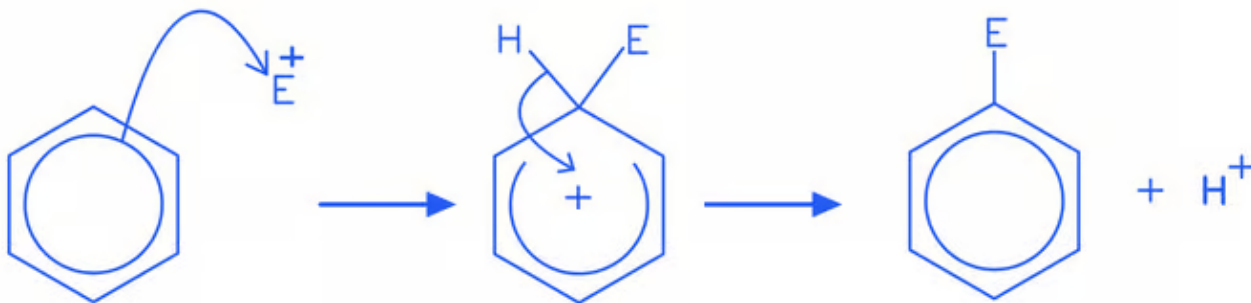


- Curly arrow from benzene ring to Br^+ ion; [1 mark]
- Positive charge in the horseshoe ring **AND** curly arrow from the middle of the C-H bond towards the positive charge of the benzene cation; [1 mark]

- Positive charge on H atom; [1 mark]

[Total: 3 marks]

- All electrophilic substitution mechanisms follow the same general pattern:



- The electrophile bonds to the benzene ring, forming an intermediate with a partially delocalised electron system – shown as a horseshoe with + inside
- There are only two locations that you should draw the + charge, either on the electron deficient carbon atom on the benzene ring or in the centre of the benzene ring
- The intermediate then loses a proton, restoring the delocalised electron system

1c

c) An equation to show the regeneration of the catalyst is:

- $\text{H}^+ + [\text{AlBr}_4]^- \rightarrow \text{HBr} + \text{AlBr}_3$; [1 mark]

[Total: 1 mark]

- $[\text{AlBr}_4]^-$ when bromine reacts with the halogen carrier, AlBr_3
- H^+ was produced when bromobenzene is produced, as shown in part (b)
- These react to regenerate the catalyst, AlBr_3
 - HBr is also produced
- The overall reaction is:
 - $\text{C}_6\text{H}_6 + \text{Br}_2 \xrightarrow{\text{AlBr}_3} \text{C}_6\text{H}_5\text{Br} + \text{HBr}$

2a

a) The reagents and conditions for each reaction are:

- Reaction 1: Cl_2 **AND** AlCl_3 / FeCl_3 / Fe / bromides of Al or Fe; [1 mark]
- Reaction 2: Cl_2 **AND** ultraviolet / UV light; [1 mark]

[Total: 2 marks]

- Depending on the reaction conditions, halogenation can occur:
 - In the aromatic ring
 - In the side chain
- Halogenation of alkylarenes in the benzene ring occurs in the presence of a halogen carrier
- AlCl_3 is the halogen carrier that you need to know, but other suitable halogen carriers are also accepted
- Halogenation of alkylarenes in the side chain will occur when the halogen is passed into boiling alkylarene in the presence of ultraviolet (UV) light

2b

b) The type of reaction mechanism that occurs in each reaction is:

- Reaction 1: Electrophilic substitution; [1 mark]
- Reaction 2: Free radical substitution; [1 mark]

[Total: 2 marks]

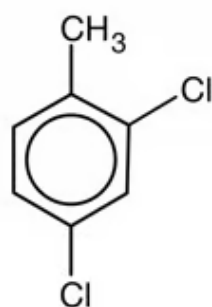
- In reaction 1, the electrophile is Cl^+ , produced by the reaction of chlorine with the halogen carrier
- In reaction 2, UV light provides the energy to break the Cl_2 homolytically, producing chlorine free radicals
- Both are substitution reactions

2c

c) The structure of an organic product and its name are:

Any **one** structure and name from the following:

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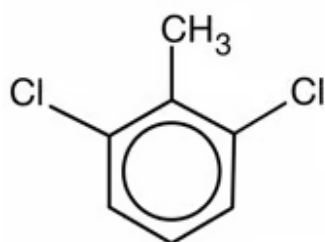


; [1 mark]

- 2,4-dichloromethylbenzene; [1 mark]

OR

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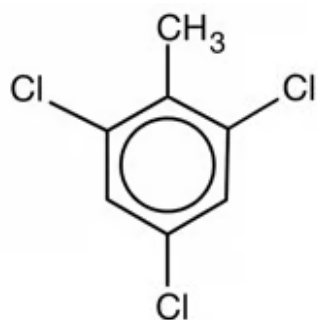


; [1 mark]

- 2,6-dichloromethylbenzene; [1 mark]

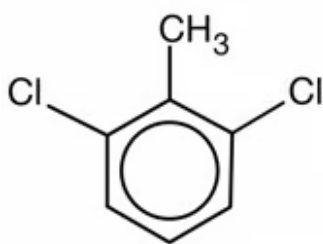
OR

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; [1 mark]

- 2,4,6-trichloromethylbenzene; [1 mark]

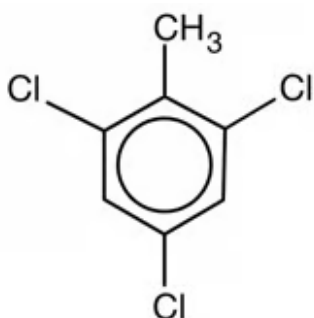


; [1 mark]

- 2,6-dichloromethylbenzene; [1 mark]

OR

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; [1 mark]

- 2,4,6-trichloromethylbenzene; [1 mark]

[Total: 2 marks]

- The methyl group directs the incoming halogen on the 2 and 4 positions, so both 2-chloromethylbenzene and 4-chloromethylbenzene are present when chloring reacts with benzene
- When in excess, chlorine can be substituted into more than one position so any of the three products above can be formed
- **Careful:** 4,6-dichloromethylbenzene is the same as 2,4-dichloromethylbenzene so is not an accepted answer

3a

a) The most reactive compound is:

- 2-bromopentane; [1 mark]

[Total: 1 mark]

- Halogenoarenes are very unreactive compared to halogenoalkanes

3b

b) The bond enthalpy of the C–Br bond in bromobenzene is greater or less than the average bond enthalpy is:

- Greater; [1 mark]
- Because the C–Br (in bromobenzene) is very strong; [1 mark]

[Total: 2 marks]

- Let the number of marks guide you to the amount of detail that is expected
- One mark is available for stating whether the bond enthalpy of the bond in bromobenzene is greater or less than the average bond enthalpy and so there is 1 mark to state what this means about the bond strength
- **Careful:** This question is not asking you to explain **why** the bond is strong, so you do not need to give a detailed explanation about the stability of the system due to the delocalisation of a lone pair of electrons on the bromine over the benzene ring

3c

c) As a hydroxide ion approaches a molecule of bromobenzene:

- Hydroxide ion / OH^- is repelled by the benzene ring; [1 mark]

[Total: 1 mark]

- The benzene ring has delocalised ring electrons above and below the carbon atoms
- The negative charge of the hydroxide ions is repelled by the benzene ring electrons
- Only in extreme conditions, such as temperatures of 200 °C and a pressure of